

## Prediction Problem

05/08/2024  
LSM

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad i=1$$

Predict  $Y$  for  $X=X_n$  (given) assuming that the model holds

$$Y_n = \beta_0 + \beta_1 X_n + \epsilon_n$$

← unobserved noise  
← random variable

Result:

$$\underset{g}{\operatorname{argmin}} \mathbb{E}(Y - g(X))^2 \text{ s.t. } \mathbb{E} g^2(X) < \infty$$
$$= \underbrace{\mathbb{E}(Y/X)}_{\text{a function of } X}$$

$$\begin{aligned} \text{Pf: } & \mathbb{E}(Y - g(X))^2 \\ &= \mathbb{E}(Y - \mathbb{E}(Y/X) + \mathbb{E}(Y/X) - g(X))^2 \\ &= \mathbb{E}(Y - \mathbb{E}(Y/X))^2 + 2\mathbb{E}((Y - \mathbb{E}(Y/X))(\mathbb{E}(Y/X) - g(X))) \\ & \quad + \mathbb{E}(\mathbb{E}(Y/X) - g(X))^2 \end{aligned}$$

$$\begin{aligned} & \mathbb{E} \left( \underbrace{\mathbb{E}[(Y - \mathbb{E}(Y/X))(\mathbb{E}(Y/X) - g(X)) | X]}_{\text{function of } X} \right) \\ &= \mathbb{E}((\mathbb{E}(Y/X) - g(X)) \underbrace{\mathbb{E}(Y - \mathbb{E}(Y/X) | X)}_0) \\ &= 0 \end{aligned}$$

## Implication for point prediction

$$Y_h = \beta_0 + \beta_1 X_h + \epsilon_h$$

Best predictor (in the sense of minimizing the mean squared prediction error) is

$$E(Y_h | X = X_h) = \beta_0 + \beta_1 X_h = \mu_h$$

Best estimator of  $\mu_h$  is  $\hat{\mu}_h = \hat{\beta}_0 + \hat{\beta}_1 X_h$

Recall For any  $0 < \alpha < 1$

$100(1-\alpha)\%$  confidence interval for  $\mu_h$  is constructed by observing that

if  $\hat{\mu}_h = \hat{\beta}_0 + \hat{\beta}_1 X_h$  then

$$\hat{\mu}_h - \mu_h \sim N\left(0, \sigma^2 \left( \frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right)\right)$$

$$\frac{\hat{\mu}_h - \mu_h}{S(\hat{\mu}_h)} = \frac{\hat{\mu}_h - \mu_h}{\sqrt{MSE \left( \frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right)}} \sim t_{n-2}$$

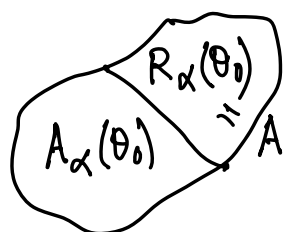
$\hat{\mu}_h \pm t_{n-2}^{1-\alpha/2} S(\hat{\mu}_h)$  is the  $100(1-\alpha)\%$  CI

## Hypothesis testing vs. Confidence interval

$P_\theta$  is a family of distribution

$X \sim P_\theta$  for some  $\theta$

Test  $H_0: \theta = \theta_0$  vs.  $H_A: \theta \neq \theta_0$  at level of significance  $\alpha$ .



$$\begin{aligned} & \chi \quad \alpha \geq \mathbb{P}_{\theta_0}(\underline{X} \in R_\alpha(\theta_0)) \\ & \Leftrightarrow 1 - \alpha \leq \mathbb{P}_{\theta_0}(\underline{X} \in A_\alpha(\theta_0)) \end{aligned}$$

In the example on estimating  $\mu_h = \beta_0 + \beta_1 X_h$

$H_0: \mu_h = \mu_h^0$  vs.  $H_A: \mu_h \neq \mu_h^0$  where  $\mu_h^0 = \beta_0^0 + \beta_1^0 X_h$

$$\frac{\hat{\mu}_h - \mu_h^0}{s(\hat{\mu}_h)} \sim t_{n-2} \quad \leftarrow \text{Acceptance region for an exact level } \alpha \text{ test}$$

$$\{(X, Y) : |\hat{\mu}_h - \mu_h^0| \leq t_{n-2}^{1-\alpha/2} s(\hat{\mu}_h)\}$$

$\hat{\mu}_h \pm t_{n-2}^{1-\alpha/2} s(\hat{\mu}_h)$  is the  $100(1-\alpha)\%$  CI

$$C_{1-\alpha}(\underline{X}) = \{\theta_0 : \underline{X} \in A_\alpha(\theta_0)\}$$

| Estimation of $\mu_h = \beta_0 + \beta_1 X_h$                              | Prediction of $Y_h = \beta_0 + \beta_1 X_h + \varepsilon_h$                    |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Point estimate<br>$\hat{\mu}_h = \hat{\beta}_0 + \hat{\beta}_1 X_h$        | Point predictor<br>$\hat{\mu}_h = \hat{\beta}_0 + \hat{\beta}_1 X_h$           |
| Interval estimate<br>$\hat{\mu}_h \pm t_{n-2}^{1-\alpha/2} s(\hat{\mu}_h)$ | Interval estimate<br>$\hat{\mu}_h \pm t_{n-2}^{1-\alpha/2} s(\hat{Y}_h - Y_h)$ |
|                                                                            | Wider interval                                                                 |

Pivotal quantity for estimating  $\mu_h$

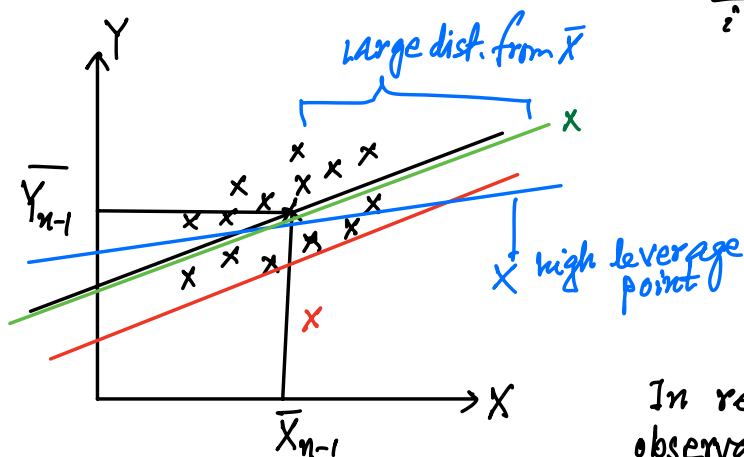
$$\frac{\hat{\mu}_h - \mu_h}{S(\hat{\mu}_h)} = \frac{\hat{\mu}_h - \mu_h}{S(\hat{\mu}_h - \mu_h)} = \frac{\text{est. error}}{S(\text{est. error})}$$

Pivotal quantity for predicting  $Y_h$

$$\frac{\hat{Y}_h - Y_h}{S(\hat{Y}_h - Y_h)} = \frac{\hat{\mu}_h - (\mu_h + \epsilon_h)}{\sqrt{\text{est Var}(\hat{\mu}_h - (\mu_h + \epsilon_h))}} \sim t_{n-2}$$

$$\begin{aligned} \text{Var}(\hat{Y}_h - Y_h) &= \text{Var}(\hat{\mu}_h - \mu_h - \epsilon_h) \\ &= \text{Var}(\hat{\mu}_h - \mu_h) + \text{Var}(\epsilon_h) \\ &= \text{Var}(\hat{\mu}_h) + \text{Var}(\epsilon_h) \\ &= \sigma^2 \left( \frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right) + \sigma^2 \end{aligned}$$

$$\therefore S^2(\hat{Y}_h - Y_h) = \text{MSE} \left( 1 + \frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right)$$



If  $X_{\text{new}} - \bar{X}_{\text{old}}$  is large, it will have a large influence on the slope of the regression line.

In regression problem,  $i^{\text{th}}$  observation (for  $1 \leq i \leq n$ ) is said to be a "high leverage point" if  $h_{ii}$  is relatively large, where  $h_{ii}$  is the diagonal element of  $H = X(X^T X)^{-1}X^T$ .

$$X = [1_n: X] \quad X^T X = \begin{bmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{n \sum X_i^2 - (\sum X_i)^2} \begin{bmatrix} \sum X_i^2 & -\sum X_i \\ -\sum X_i & n \end{bmatrix}$$

Ex: Compute value of  $h_{ii}$ .

Note:  $\underline{e} = \underline{Y} - \hat{\underline{Y}} = (I - H) \underline{Y} \stackrel{\substack{\uparrow \\ \text{assuming regression} \\ \text{model to be true}}}{=} (I - H) \underline{\epsilon}$

$$\begin{aligned} \text{Var } \underline{e} &= \sigma^2 (I - H)^2 \\ &= \sigma^2 (I - H) \end{aligned}$$

$$\text{Var}(e_i) = \sigma^2 (1 - h_{ii})$$

Question: How large can it be

$$\underline{\underline{1_2}} \lambda_{\max} = \sup_{\|a\|=1} a^T H a \geq e_i^T H e_i \rightarrow i^{\text{th}} \text{ diagonal elt } h_{ii}$$

Ex: When can  $h_{ii}$  be equal to 1?

Average is  $\frac{2}{n}$  (since e.v. of  $H$  is  $\underbrace{\frac{1}{2}}_{\frac{1}{2}}, \underbrace{0}_{n-2}$ )

Quantification of the influence of one observation on the slope of the regression line

Let  $\hat{\beta}_0^{(n-1)}$  and  $\hat{\beta}_1^{(n-1)}$  be the OLS est. of  $(\beta_0, \beta_1)$  based on  $\{(X_i, Y_i)\}_{i=1}^{n-1}$ . Let  $(\hat{\beta}_0, \hat{\beta}_1)$  be the fitted regression coefficient based on the whole data

then define  $e_n^{(n-1)} = Y_n - (\hat{\beta}_0^{(n-1)} + \hat{\beta}_1^{(n-1)} X_n) =$  predictor error for  $Y_n$  based on remaining obs.

$$\hat{\beta}_1 = \hat{\beta}_1^{(n-1)} + \frac{(1 - \frac{1}{n})(X_n - \bar{X}_{(n-1)})}{\sum_{i=1}^n (X_i - \bar{X})^2} e_n^{(n-1)}$$

$$= \hat{\beta}_1^{(n-1)} + \frac{(X_n - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2} e_n^{(n-1)} \quad (\text{Exc.})$$